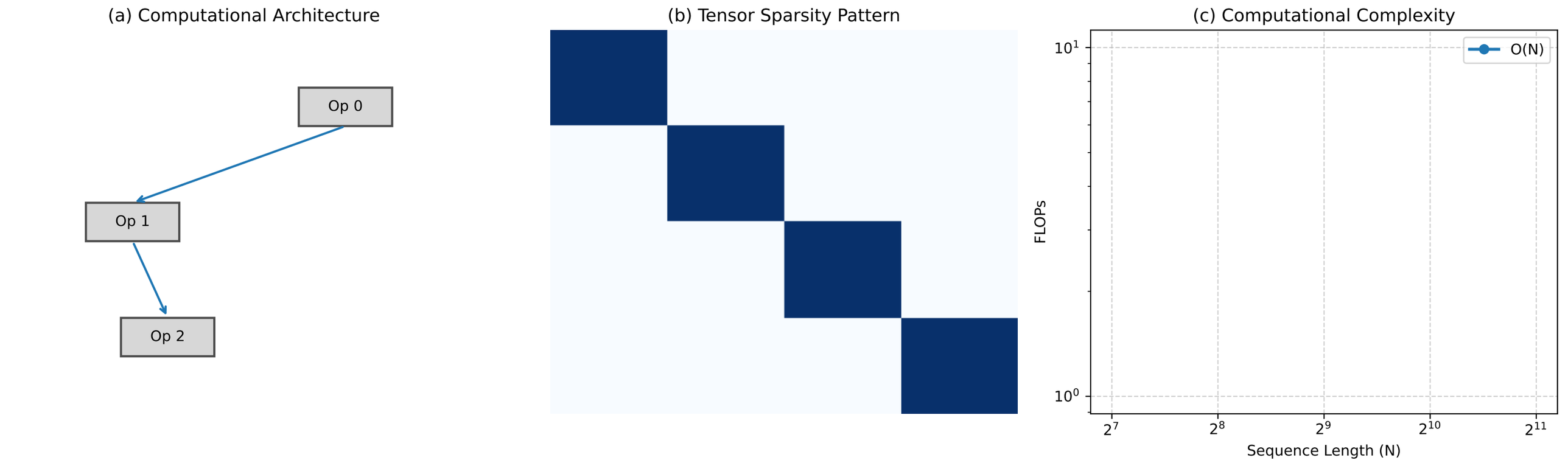Module Architecture: partitioner